Artemis Track 1

Systematic Crypto Factor Rebalancing Strategy

Cross-Sectional Factor Portfolio with Weekly Rebalancing

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Research Period: 2017–2026

In-Sample: 2021-05-10 – 2024-11-11

Out-of-Sample: 2024-11-18 – 2026-05-25

79 weeks out-of-sample – ∼113 Crypto Assets

This report presents a systematic crypto factor rebalancing strategy for Track 1 of the Artemis Quant Competition.

The goal is a rules-based portfolio with an economic rationale, statistical defensibility, and honest evaluation of limitations.

Contents

1. Executive Summary

This report presents a systematic crypto factor rebalancing strategy for Track 1 of the Artemis Quant Competition.[1] The objective is to construct a rules-based portfolio that can be explained economically, tested statistically, and rebalanced on a **fixed weekly schedule**.[2]

The core idea is simple: identify measurable characteristics of crypto assets that predict relative returns, combine them into a portfolio, and rebalance weekly. The implementation, however, requires careful attention to multiple-testing discipline, regime-aware risk sizing, and honest assessment of what the data actually supports versus what it merely suggests.

Headline Result

The **Sharpe Ensemble** achieves an out-of-sample Sharpe ratio of +0.84 (annualized return +29.9%, volatility 35.8%, max drawdown −24.7%) across 79 weeks of held-out data from November 2024 to May 2026. Bitcoin returned −12.3% annualized over the same period, and the equal-weight market returned −34.4% annualized.

However, honest decomposition shows that this Sharpe ratio depends heavily on a single composite factor (MispricingM) and that the Priced Tilt sub-book is a net drag out-of-sample. The strategy is therefore best described as a *single-factor portfolio with volatility and drawdown controls*, not a genuinely diversified multi-factor book. All limitations are discussed openly.

2. Glossary

Term	Definition	Key Reference
Factor	A rule that ranks or tilts assets using a characteristic such as volatility, size, momentum, or drawdown	[[7]], [[13]]
Rebalancing	Periodically updating portfolio weights; this project rebalances weekly	[[6]]
IC	Information coefficient: the rank correlation between a factor score and next-period returns	[[21]], [[11]]
Newey-West t	A significance measure adjusted for autocorrelation and heteroskedasticity	[[19]]
ASD	Almost stochastic dominance: a distributional test suited to non-normal returns	[[15]], [[13]]
GX pricing	Giglio-Xiu latent-factor pricing: tests risk premia controlling for hidden common factors	[[10]]
Sharpe ratio	Return per unit of volatility	[[20]]

[1] A *factor* is a measurable asset characteristic used to predict or explain cross-sectional returns. The concept is central to modern asset-pricing research.[[7]], [[4]]

[2] *Out-of-sample* refers to a time period reserved for testing after all design choices have been made. This is the standard method for reducing overfitting risk, though a single holdout window remains a limited test.

Term	Definition	Key Reference
Max drawdown	Largest peak-to-trough portfolio loss during the test window	[[17]]
XGBoost	Gradient-boosted tree model used here to estimate regime probabilities	[[5]]
Risk premium	Extra expected return for bearing systematic risk	[[7]]
Latent factor	A hidden common risk extracted statistically from returns	[[10]]
Overfitting	Fitting historical noise rather than a repeatable relationship	[[5]], [[3]]
Survivorship bias	Data problem where failed or delisted assets are missing from the historical universe	[[13]], [[3]]
TVL	Total value locked: a DeFi protocol engagement metric sourced from Artemis Terminal; strongly priced as a systematic risk but not a reliable standalone weekly trading signal	[[2]], [[3]]
VolC	Volatility characteristic: a low-volatility crypto ranker	[[9]]
MAXRET	Maximum recent return: a lottery-reversal signal	[[1]], [[13]]
RMOM	Risk-adjusted momentum: recent return scaled by recent volatility	[[13]]
SMBC	Small-minus-big crypto size factor	[[7]], [[3]]
NetRel	Network-relative signal comparing a coin with related coins	[[12]]
CRASH8	Rebound signal for deeply crashed coins	[[13]]
BETA26	High-beta crypto risk tilt	[[9]], [[3]]
SKEW52	Skewness or lottery-like payoff tilt	[[1]]
NEWC	Newer-coin seasoning tilt	[[13]], [[3]]

The report uses **in-sample** to mean the research window used to study and build the model, and **out-of-sample** to mean the later window used to test whether the model remains effective after design choices have been finalized.

3. Competition Objective

Track 1 asks for a systematic crypto factor rebalancing strategy. In practical terms, this means:

1. Define a tradable crypto universe.
2. Build factor signals with a clear economic story.
3. Rebalance the portfolio on a consistent weekly schedule.
4. Evaluate returns, risk, turnover, drawdown, and robustness.
5. Critically explain what could fail.

The following sections address each requirement in turn, proceeding from data and universe construction through factor discovery and validation, strategy assembly, backtest results, and finally an honest critical evaluation.

4. Universe, Data, and Design Choices

The research uses a weekly crypto asset panel covering approximately 113 large-cap and upper-mid-cap cryptocurrency assets. The broad factor validation work spans 2017–2026, with the final strategy evaluated on a panel whose last 79 weeks constitute the out-of-sample window ending on May 25, 2026.

Split	Period
In-sample (main validation)	2021-05-10 – 2024-11-11
Out-of-sample (main validation)	2024-11-18 – 2026-05-25
In-sample (final strategy)	Before 2024-11-25
Out-of-sample (final strategy)	Final 79 weeks

The universe is drawn from CoinGecko market-capitalization rankings, with deliberate exclusions of stablecoins, wrapped duplicates, and bridged tokens. Price, market-capitalization, and volume data are sourced from CoinGecko; on-chain metrics including protocol fee revenue and total value locked are sourced from the Artemis Terminal data API. This universe occupies a specific economic niche that balances three competing considerations.

4.1 Why This Universe?

Efficiency and coverage gaps. The largest crypto assets (top 20 by market cap) are subject to intensive analyst coverage, institutional arbitrage via ETF flows, and deep on-chain surveillance. Cross-sectional factor premia in these names are likely compressed by the same mechanism that erodes anomalies in large-cap equities.[[18]] By extending into the upper-mid-cap tier (#20–#120), the universe captures assets where information asymmetry is structurally higher—fewer dedicated analysts, thinner institutional ownership, more heterogeneous investor bases—precisely the conditions under which behavioral mispricings persist.

Execution feasibility. Extending below the top 120 into micro-cap crypto introduces severe capacity constraints. Daily trading volumes below $1 M create market-impact costs exceeding 100 bps per trade, rendering paper factor premia untradeable at any meaningful scale. The chosen universe ensures that most constituents trade above $5 M daily volume on major exchanges, keeping estimated one-way execution costs in the 10–30 bps range for institutional-sized orders.

Cross-sectional dispersion. Factor strategies require sufficient return dispersion across assets to generate meaningful long-short spreads. A universe restricted to the top 10 assets offers too little variation in characteristics like size, volatility, and momentum. The ∼113-asset panel provides enough dispersion for quantile-sorted portfolios (top/bottom 30%) to contain 15–20 assets per leg, ensuring adequate diversification within each portfolio tail.

4.2 Validation Period and Dynamic Market Cap

The primary validation window is anchored in May 2021 to leverage a robust five-year data history while enforcing strict liquidity and survivorship controls. Assets are restricted to a minimum market capitalization of $100 M; any coin falling below this threshold is dynamically excluded from the following week's eligible list and its positions are immediately closed.

Crucially, this five-year timeline captures the profound post-COVID structural transformation of the crypto asset class. By training the strategy through the retail-driven speculative excess of 2021 and the systemic contagion of 2022, and validating it against the institutionalized, ETF-driven landscape of 2024–2026, we ensure that our factors demonstrate resilience across both historical liquidity crises and a permanently altered market structure.

The strategy excludes stablecoins, wrapped duplicates, and very low-quality assets wherever possible. Stablecoins lack the risk-return profile of volatile crypto assets. Wrapped assets duplicate exposures already present in the universe. Very illiquid assets produce paper profits that may not be tradable after accounting for slippage.

5. Factor Discovery and Validation Framework

The strategy begins with factor discovery. A factor is useful only if it passes at least one of three independent tests, each answering a different economic question. A factor does not need to pass all three: a 264-week crypto panel is too short to deliver $|t| \geq 2$ on every axis for every factor.

5.1 Test 1 — Information Coefficient (IC)

The information coefficient is the Spearman rank correlation between a factor's cross-sectional scores at week t and the realized returns at week $t{+}1$. A positive direction-adjusted IC means the factor's bet ranked next-week winners above next-week losers. Typical useful ICs in crypto are small ($|\text{IC}| \approx 0.03$–0.05) because the asset class is noisy, but even small ICs compound into meaningful alpha when applied consistently across 50+ assets each week.

An important subtlety: some factors go *long the bottom* of their characteristic sort. VolC, for example, buys low-volatility coins, so its raw IC against the volatility characteristic is *negative*—which is the factor *working*, not failing. All IC results are reported direction-raw; the sign should be read against the factor's intended bet direction. Statistical significance is assessed via Newey-West t-statistics with 4 lags, which correct for the serial correlation and heteroskedasticity endemic in weekly crypto return series.[[19]]

5.2 Test 2 — Almost Stochastic Dominance (ASD)

The Sharpe ratio implicitly assumes normally distributed returns. Crypto factor returns are not: every factor in our panel rejects the Jarque-Bera normality test at the 5% level, with excess kurtosis ranging from +1.1 to +19.3 and skewness from −1.0 to +3.1. Under these conditions, the Sharpe ratio understates tail risk and can rank strategies incorrectly.

Almost stochastic dominance avoids this by comparing entire return distributions without parametric assumptions.[[15]] The test produces two violation measures:

- ε_1 (AFSD): if $\varepsilon_1 \leq 5.9\%$, the factor *almost first-order stochastically dominates* Bitcoin—meaning *all* investors, regardless of risk preference, would prefer the factor's return distribution.
- ε_2 (ASSD): if $\varepsilon_2 \leq 3.2\%$, the factor *almost second-order dominates* Bitcoin—meaning all *risk-averse* investors would prefer it.

Recent crypto factor work applies this framework to cryptocurrency factor portfolios and demonstrates that several factors dominate benchmarks under ASD even when their Sharpe ratios are unremarkable, making ASD the appropriate test for fat-tailed crypto returns.[[13]]

5.3 Test 3 — Giglio-Xiu Latent-Factor Pricing (GX)

The first two tests ask whether a factor *works as a trading signal*. The Giglio-Xiu test asks a fundamentally different question: is the factor a *priced source of systematic risk* in the cross-section? A factor is "priced" if assets that load more heavily on it earn systematically different expected returns—a risk premium, denoted λ.[[10]]

Three pricing methods are run side by side to triangulate the evidence:

1. **Fama-MacBeth (FMB)**: week-by-week cross-sectional regressions of returns on factor betas, averaging the estimated risk premia λ_t across weeks with Newey-West standard errors. This is the standard approach[[8]] but is noisy when the number of factors approaches the number of assets per week.
2. **GX observed-only**: a single cross-sectional regression on time-series average returns using heteroskedasticity-robust standard errors. More stable than FMB but ignores the possibility that omitted common risks contaminate the estimates.
3. **GX full**: the most credible specification.[[10]] After initial pricing, Bai-Ng information criteria select K hidden latent factors from the pricing residuals. Betas are then re-estimated on the combined observed + hidden factor set, and risk premia are re-priced. This removes contamination from unobserved systematic forces that would otherwise inflate or deflate the estimated λ for the observed factors.

The GX-full t-statistic is the primary pricing metric reported throughout. A factor with $|t_{\text{gx}}| \geq 1.65$ is considered a statistically meaningful priced risk.

5.4 From Tests to Portfolio Routes

The three tests are not averaged into one score. They answer different questions, so they route factors to different jobs:

1. **Economic story required**: a factor needs a plausible mechanism before it is tested.
2. **IC route**: factors with robust in-sample and out-of-sample ranking power become weekly rankers.
3. **ASD route**: factors whose return distributions dominate Bitcoin become distributional mispricing candidates.
4. **GX route**: factors that remain priced after observed and hidden risks are controlled for become priced-risk candidates.
5. **Control route**: market, sparse-PCA, and macro-spanned directions are retained as pricing controls, not alpha trades.

This is the central design choice. A weak IC does not automatically eliminate a factor, and a strong GX statistic does not automatically justify a large allocation. The test that a factor passes determines the job it is allowed to perform.

6. Factor Economics and Evidence

6.1 The Economic Question

Every factor starts with the same question: why would this trade pay? The answer cannot be "because the backtest liked it." A factor is a repeated long–short trade, and the economic story determines both the direction of the trade and the test it should be judged by. The factors fall into five families.

6.2 Classic Risk and Anomaly Factors

Factor	**Trade**	**Why it might pay**
RC	Hold the whole crypto market, value-weighted	This is the crypto market premium: compensation for bearing systematic crypto risk. It is not alpha because every long-only holder already earns it. We include RC so the other factors are measured net of market beta.
SMBC	Long small coins, short big coins	Small caps are less liquid, less covered, and riskier, so they should require extra expected return. This is the crypto version of the Fama–French SMB effect, though it can invert during "flight to quality" periods when capital crowds into BTC and ETH.[[7]]
MomC	Long 4-week winners, short losers	Investors can under-react to news, so recent winners may keep winning.[[14]] In crypto, raw momentum also crashes violently when trends reverse.
VolC	Long calm coins, short volatile coins	Leverage-constrained and lottery-seeking investors can overpay for exciting high-volatility names, leaving lower-volatility coins cheap. This is the crypto analogue of betting against beta.[[9]]

6.3 Risk-Adjusted Momentum and Lottery Reversal

Risk-adjusted momentum divides a recent trend by the coin's own recent volatility, which makes the signal closer to a short-horizon Sharpe ratio. The purpose is to strip out the noise that makes raw momentum unstable in crypto.

Factor	Trade
RMOM1w	Long coins with high 1-week return divided by 4-week volatility.
RMOM2w	Long coins with high 2-week return divided by 4-week volatility.
RMOM4w	Long coins with high 4-week return divided by 4-week volatility.
MAXRET	Rank by the highest single weekly return in the last 4 weeks. An extreme up-week is a lottery signal: if investors overpay for the jackpot coin, the correct trade is to fade the lottery.[[1]], [[13]]

6.4 Fundamental Factors

Factor	Trade	Why it might pay
FunC	Long high fees-to-market-cap, short low	This is crypto value: protocols that generate real cash flow should be cheap relative to protocols that do not. Fee revenue data is sourced from Artemis Terminal. The problem is coverage: only about half the universe earns meaningful fees, so the factor is structurally under-powered.
TVLC	Long high TVL-to-market-cap, short low	This captures DeFi engagement per dollar of market value, using TVL data from Artemis Terminal. The bull case is usage-backed value; the bear case is TVL irrelevance because the information is already priced. The sign is genuinely ambiguous before testing.[[2]]

TVLC illustrates a pattern that recurs across the factor set: being strongly priced is not the same as being a good weekly trading signal. The Giglio–Xiu test assigns TVLC a large negative risk premium ($\lambda = -185.2\%/\text{yr}$, $|t| = 4.71$), meaning high-TVL/market-cap assets earn *lower* expected returns — investors appear to pay a DeFi engagement premium, leaving low-TVL protocols underpriced on a risk-adjusted basis. Yet TVLC has no measurable weekly IC. The Artemis Terminal data confirms the usage signal is real; the market simply prices it in at a slow, structural pace rather than week to week. This is why TVLC is routed to the Priced Tilt sleeve rather than the weekly ranker book.

6.5 Network and Structure Directions

Network and structure factors need more unpacking because they are not all buy/sell signals. Some describe how crypto assets move in blocs. These directions are useful for diversification and for pricing controls, but many are not alpha trades.

Coin networks. A network is a set of dots connected by lines. Here each dot is a coin and each line means that two coins move together. We build the network from returns rather than drawing it by hand:

1. **Correlate every pair of coins.** On each rolling 12-week window, compute the Spearman rank correlation between every pair of coin returns. Spearman correlation is used because crypto has large outliers and rank correlation is less distorted by one extreme week.[[21]]

2. **Turn correlation into distance.** Distance is $\sqrt{2(1-\rho)}$. Correlation $+1$ becomes distance 0, while unrelated coins sit farther apart.
3. **Keep the strongest skeleton.** A full graph connects every coin to every other coin and is too noisy. A minimum spanning tree keeps the smallest set of links needed to connect the universe, leaving the backbone of strongest relationships.
4. **Find communities.** Louvain clustering groups the tree into communities where coins are more connected to each other than to outsiders. The clusters often resemble crypto narratives, but the algorithm only sees co-movement.

Once clusters exist, two network factors follow naturally. NetMom is within-cluster momentum: inside each cluster, it compares each coin's 4-week momentum with its clustermates and backs the local leader against the laggard. NetRel is cross-cluster rotation: it compares each coin or cluster against the rest of the market and backs the narrative bloc pulling ahead of the others. This follows the broader idea that crypto returns contain asset-specific and network-like components.[[16]], [[12]]

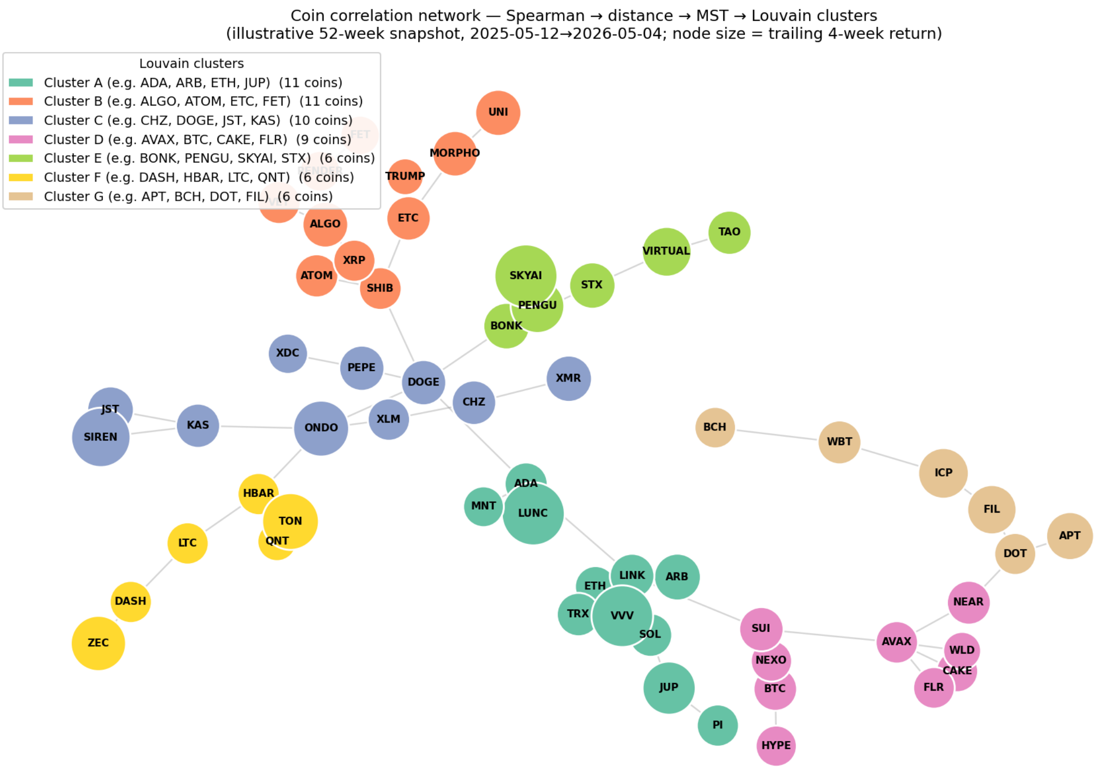

Figure 1. Coin correlation network with Louvain clusters. The graph is built from this project's own return panel using the Spearman correlation, distance transform, minimum spanning tree, and Louvain community detection pipeline. Each dot is a coin, each line is a strong retained co-movement link, and each colour is a data-driven cluster.

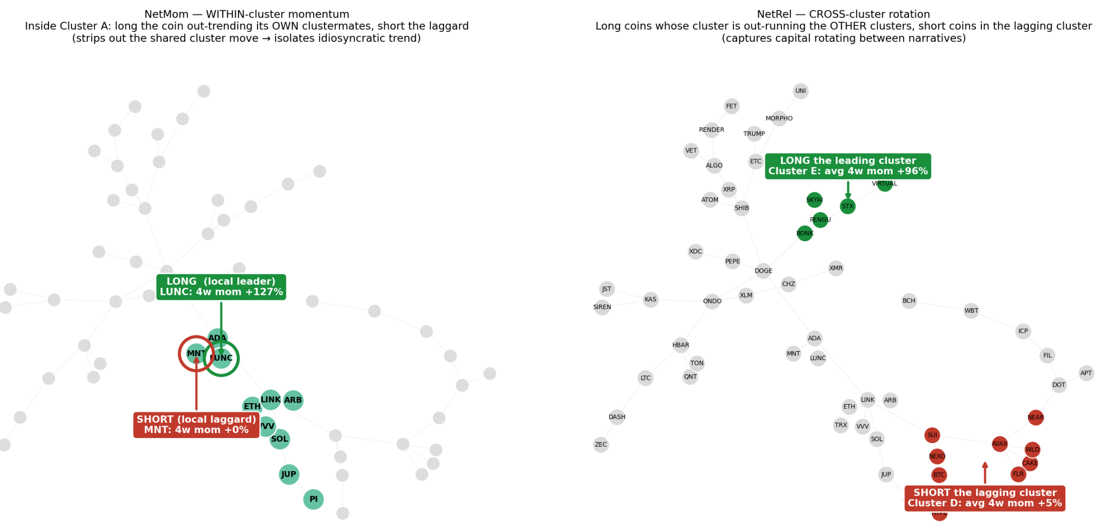

Figure 2. NetMom and NetRel illustrated on the same network. NetMom works inside one cluster by buying the local momentum leader and shorting the local laggard. NetRel works between clusters by buying coins in the leading narrative bloc and shorting coins in the lagging bloc. The figures are produced by `figures/make_network_viz.py`.

Sparse PCA. Ordinary PCA finds the main directions along which all coins move, but the resulting components load a little on many coins and are hard to interpret. Sparse PCA adds a penalty that pushes most loadings to zero, so each component loads on a smaller and more readable group. We extract four sparse components: SPC1 is a broad DeFi-majors or "everything together" direction; SPC2 is a payment and old-guard direction; SPC3 is an alt-L1 direction; and SPC4 is a legacy, privacy, and exchange-token direction. These are market-structure directions, not standalone trades.

CCA. Canonical correlation analysis asks how much of crypto is macro in disguise. One side contains the crypto return panel; the other contains BTC, SPX, DXY, UST10Y, and VIX. CCA finds the crypto-side combinations most correlated with macro-side combinations. CCA1–CCA3 therefore measure the slices of crypto that move with equity risk appetite, the dollar, rates, and volatility. They are controls, not alpha bets.

Direction	What it represents	How to use it
NetMom	Within-cluster relative momentum	A coin-level signal: buy the local winner and short the local laggard inside the same narrative cluster. It removes the shared cluster move and tests idiosyncratic trend.
NetRel	Cross-cluster narrative rotation	A rotation signal: buy coins in clusters pulling ahead of the rest of crypto and short coins in lagging clusters. It captures capital moving between narratives.
SPC1–SPC4	Sparse crypto-bloc directions	Risk controls: each component is a readable co-movement bloc such as broad market/DeFi-majors, payments, alt-L1s, or legacy/exchange exposure.
CCA1–CCA3	Macro-spanned crypto directions	Macro controls: these measure the part of crypto that moves with BTC, equities, the dollar, rates, and volatility. They prevent macro beta from being mistaken for crypto alpha.

Table 4. Summary of network and structure directions.

6.6 Behavioral Factors

The behavioral factor search tested 182 candidate specifications, then kept only factors that were economically interpretable, distinct from renamed VolC/MAXRET variants, and robust to multiple-testing correction.

Factor	Trade	Investor behavior captured
NEWC	Long young coins, short seasoned coins	Newness or seasoning premium: investors demand compensation for holding less-proven assets.
SKEW52	Long high 52-week skew, short low	Lottery memory: investors remember salient right-tail jumps and overpay for coins with upside histories.
CRASH8	Long recently crashed coins, short resilient coins	Capitulation premium: freshly punished coins carry a priced rebound or crash-risk exposure.
BETA26	Long high 26-week beta, short low	Speculative risk-on demand: high-beta coins are what investors reach for when they want amplified crypto exposure.

6.7 Master Evidence Table

The table below puts every factor through the same evidence lenses. IC t-statistics measure weekly ranking power; ASSD marks whether the factor return distribution is preferred to Bitcoin by risk-averse investors; GX reports the full Giglio–Xiu priced-risk test after observed and hidden factors are controlled for.[[10]]

#	Factor	Family	Economic one-liner	IC IS t	IC OOS t	ASD	GX λ/yr	GX t	Grade
1	VolC	Anomaly	Low-vol beats high-vol	−3.32	−4.31	No	−185.8%	−5.05	Confirmed
2	MAXRET	Lottery	Short the lottery max-return signal	−3.49	−4.05	No	+160.0%	+5.52	Confirmed

#	Factor	Family	Economic one-liner	IC IS t	IC OOS t	ASD	GX λ/yr	GX t	Grade
3	CRASH8*	Behavioral	Capitulation rebound	–	–	–	+177.9%	+4.79	Behavioral-confirmed
4	BETA26*	Behavioral	Speculative risk-on demand	–	–	–	+122.8%	+4.60	Behavioral-confirmed
5	TVLC	Fundamental	DeFi engagement, TVL/mcap	–	–	–	−185.2%	−4.71	Priced risk
6	RC	Market	Crypto market premium	–	–	–	+217.2%	+3.92	Priced risk
7	SKEW52*	Behavioral	Lottery memory, skew	–	–	–	+96.9%	+3.44	Behavioral-confirmed
8	NEWC*	Behavioral	Newness and seasoning	–	–	–	+114.1%	+3.43	Behavioral-confirmed
9	RMOM1w	Risk-adj mom	1-week Sharpe momentum	−1.07	−0.27	Yes	+59.4%	+1.86	Priced risk
10	SMBC	Anomaly	Size, small minus big	+1.49	+1.25	Yes	+9.6%	+0.36	Suggestive
11	NetRel	Structure	Cross-cluster rotation	−1.72	−0.10	Yes	−0.2%	−0.01	Suggestive
12	RMOM2w	Risk-adj mom	2-week Sharpe momentum	−2.46	−0.24	Yes	+31.3%	+1.05	Suggestive
13	Mispricing M	Composite	Equal-weight ASSD-dominant blend	–	–	Yes	–	–	Suggestive
14	FunC	Fundamental	Fees/mcap value	–	–	–	+6.6%	+0.40	Economic-only
15	RMOM4w	Risk-adj mom	4-week Sharpe momentum	−2.61	−0.86	No	+36.6%	+1.24	Not supported
16	MomC	Anomaly	Raw 4-week momentum	−1.66	+0.13	No	−3.3%	−0.10	Not supported
17	NetMom	Structure	Within-cluster momentum	−0.15	−0.81	No	+8.6%	+0.37	Not supported
18	SPC3	Structure	Alt-L1 bloc direction	–	–	No	−221.4%	−3.88	Structure
19	SPC1	Structure	DeFi-majors bloc direction	–	–	No	+57.6%	+1.08	Structure
20	SPC2	Structure	Payment/old-guard direction	–	–	No	−56.0%	−1.47	Structure
21	SPC4	Structure	Legacy/exchange direction	–	–	No	+93.5%	+1.42	Structure
22	CCA1	Structure	Macro-spanned direction 1	–	–	–	−55.0%	−0.58	Structure
23	CCA2	Structure	Macro-spanned direction 2	–	–	–	+6.6%	+0.58	Structure
24	CCA3	Structure	Macro-spanned direction 3	–	–	–	+14.9%	+0.58	Structure

*Behavioral factor priced jointly with $K_{\text{hidden}} = 2$ hidden factors in the behavioral pricing run. Base factors use the full-sample GX-full pricing run.

6.8 How to Read the Grades

Grade	Meaning
Confirmed	Strong on multiple independent lenses; defensible core signal.
Behavioral-confirmed	New behavioral factor that is strongly GX-priced and survives Bonferroni multiple-testing correction.
Priced risk	Compensated systematic exposure, but not necessarily a week-to-week trading edge.
Suggestive	Economic story remains intact with partial or semi-significant evidence.
Economic-only	Sound rationale, but this data cannot confirm it.
Structure	Risk direction describing how the market moves, not an alpha bet.
Not supported	Fails its own prediction on this sample.

The grading is deliberately not pass/fail. A 264-week crypto panel cannot hit $|t| \geq 2$ everywhere, and a factor can be real on one lens while silent on another. VolC and MAXRET show the key nuance. Their weekly ranking evidence is strong: low-vol wins week to week, and lottery spikes reverse. But their long-run priced-premium evidence can point in the opposite direction from what a pure alpha-signal interpretation would suggest. The short-term ranking edge and the multi-year risk premium are not the same trade. That is why the strategy routes factors to different jobs instead of mechanically trading one master score.

6.9 Factor Validation: Selected Exhibits

The three charts below illustrate cumulative long-short returns and rolling IC significance for three representative factors spanning different roles in the strategy: VolC (confirmed weekly ranker), CRASH8 (behavioral priced-risk factor), and NetRel (structure/ASD route).

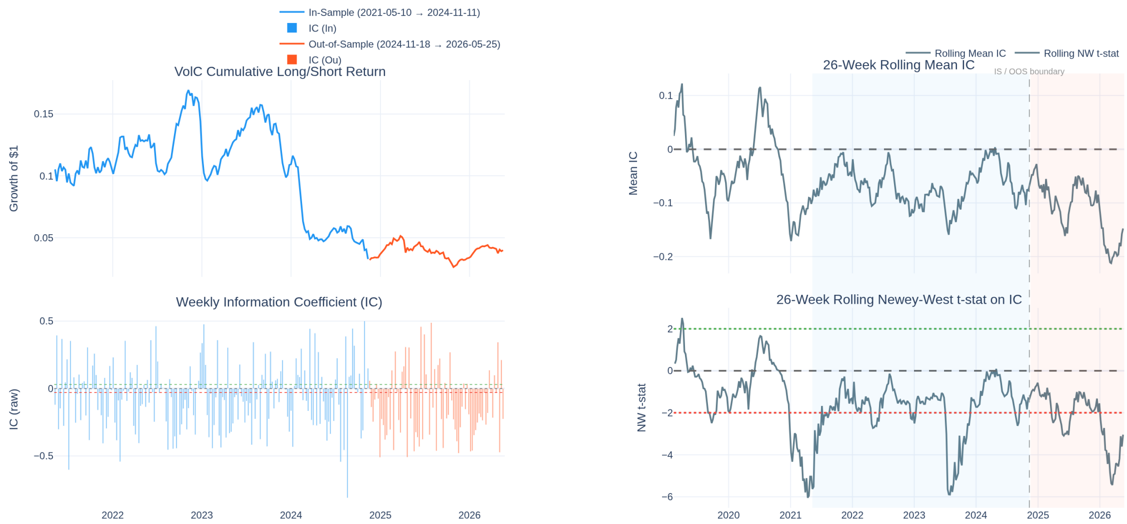

Figure 3. VolC (low-volatility anomaly). Left: cumulative long-short return with IS/OOS split. Right: rolling Newey-West IC t-statistic; values outside the ± 1.65 band indicate significant weekly ranking power.

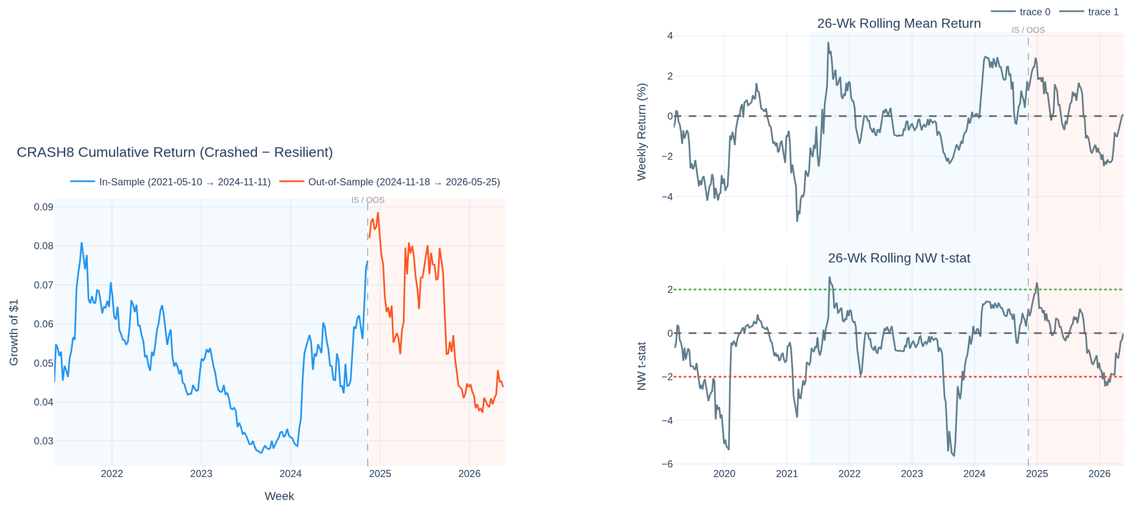

Figure 4. CRASH8 (capitulation rebound). Left: cumulative long-short return. Right: rolling return significance. CRASH8 is priced by the GX test ($t = +4.79$) rather than a consistent IC signal, so significance is episodic and concentrated around drawdown events.

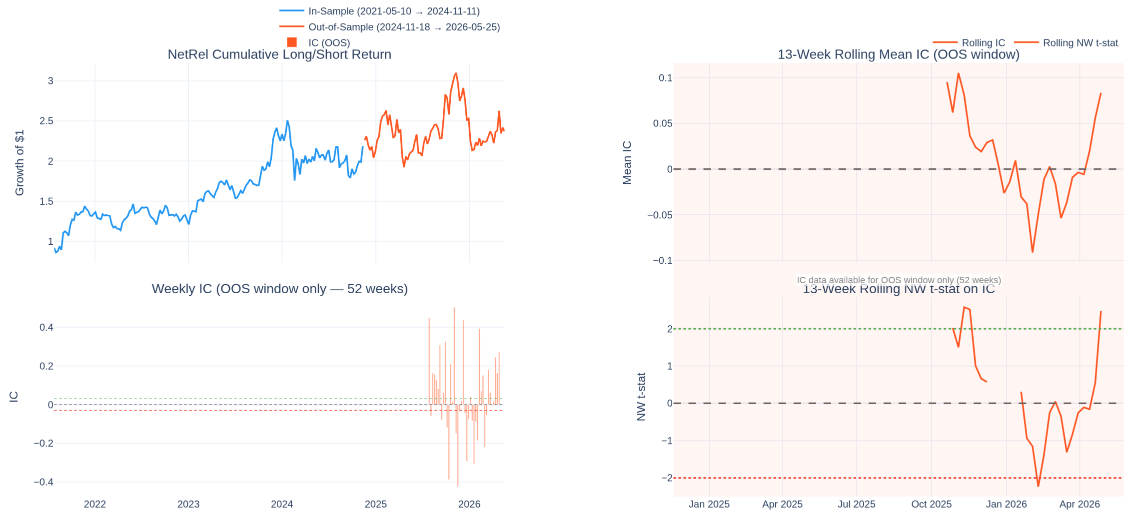

Figure 5. NetRel (cross-cluster narrative rotation). Left: cumulative long-short return. Right: rolling IC significance. NetRel's IC is individually weak ($t = -0.10$ OOS) but its return distribution almost-stochastically dominates Bitcoin, justifying its role in the MispricingM composite rather than as a standalone ranker.

7. Final Strategy Construction

The key design choice in the final strategy construction is *not* to blend every good-looking factor into one composite score. A single score is tempting because it is simple, but it erases the main lesson from Section 5: factors are real in different ways. Some are weekly rankers, some are distributional mispricing edges, and some are priced systematic exposures. The final strategy therefore builds three separate sub-books, where each book groups factors that came out of the same exit of the factor router.

7.1 From Router to Books

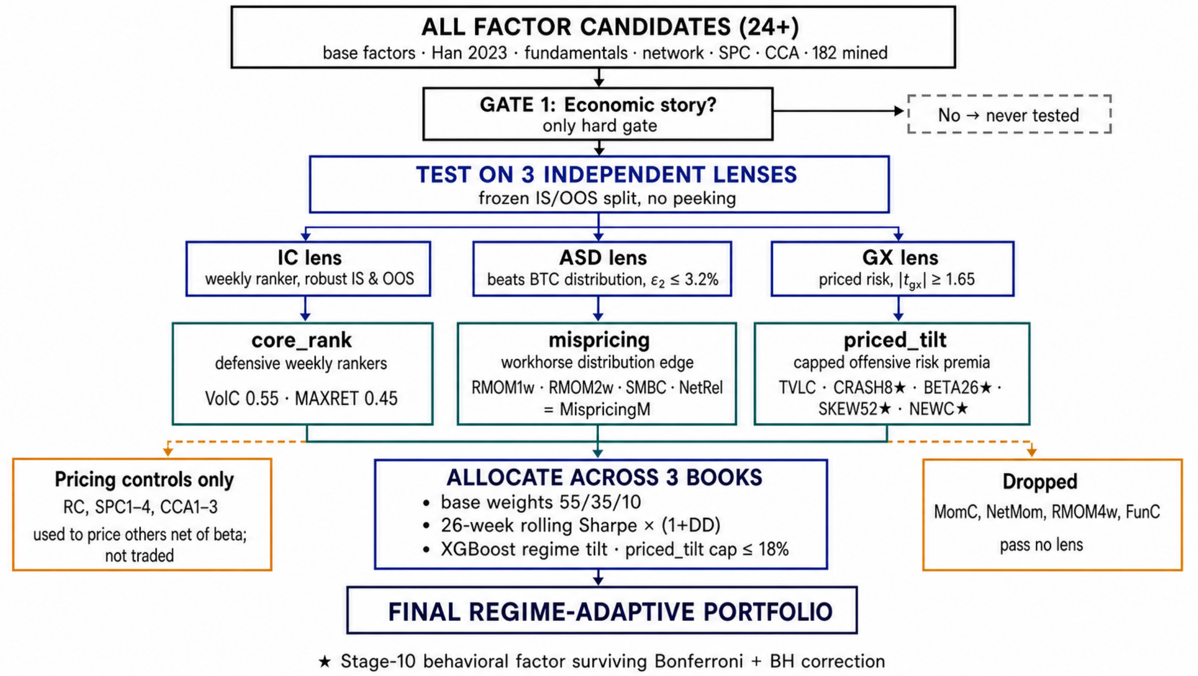

Figure 6. Factor routing graph used to construct the three strategy books. The IC route becomes Core Rank, the ASD route becomes MispricingM, and the GX route becomes Priced Tilt. This preserves the kind of evidence each factor has instead of forcing all factors into one master score.

Sub-book	Members and weights	What unifies them	When it works
core_rank	VolC 0.55; MAXRET 0.45	Confirmed IC-robust weekly rankers.	Defensive sleeve for risk-off markets; low-volatility and lottery-fade act like flight-to-safety trades.
mispricing	RMOM1w, RMOM2w, SMBC, NetRel; 0.25 each	ASSD-dominant distribution edges; the MispricingM composite.	Workhorse alpha sleeve for normal markets; trimmed modestly in risk-off regimes.
priced_tilt	TVLC 0.25; CRASH8 0.20; BETA26 0.20; SKEW52 0.20; NEWC 0.15	Priced behavioral and DeFi systematic exposures.	Offensive sleeve for risk-on markets; capped because it is newest and most data-mined.

Strategy sub-books and their economic roles.

There are three reasons to keep these books separate rather than combine them into one signal.

1. **Factors of the same kind share a regime.** Weekly rankers work best when the market is

defensive; priced and behavioral tilts are naturally more offensive. A monolithic score cannot lean toward the sleeve that matches the current market state.

2. **Within-book averaging diversifies common noise.** The four mispricing sleeves are individually only Suggestive, but they share a common mispricing signal; averaging keeps the common signal and cancels idiosyncratic noise.[[22]], [[13]] The behavioral sleeve is also designed to avoid putting all weight on one correlated effect.
3. **Different books need different risk budgets.** A defensive ranker book can carry more gross exposure in a downturn; an offensive tilt book should be cut hard in the same environment. Separate books let each sleeve carry its own regime-dependent gross exposure.

7.2 Why MispricingM Becomes a Book

The mispricing book is exactly the MispricingM composite developed in Section 5: SMBC, NetRel, RMOM1w, and RMOM2w at equal weights. Individually, these factors are not clean weekly rankers. Together, they form the equal-weight blend that almost-stochastically dominates Bitcoin. The final strategy construction simply promotes that research prototype into a live sub-book.

Window	Ann. return	Sharpe	t-stat
IS	+31.0%	+1.23	+2.31
OOS	+34.1%	+1.50	+1.60

MispricingM performance by window.

This is why NetRel reappears in the final strategy. It is not being smuggled back in after failing as a weekly ranker. It was routed here by the ASD lens in Section 5, and this book is the trading home of that route.

The honesty flag is kept visible: factor selection for the mispricing book uses full-sample ASD, which technically sees the OOS window. To make that transparent, the validation pipeline reports ε_2 separately for IS, full sample, and OOS. The dominance is not only an OOS artifact, but the selection process is still not a perfectly untouched holdout.

7.3 Why Priced Tilt Is Low-Correlation

The priced_tilt book is deliberately not one behavioral trade repeated under four names. The behavioral factors are chosen to be economically distinct and mostly low-correlation:

	NEWC	SKEW52	CRASH8	BETA26
NEWC	+1.00	+0.03	+0.63	+0.10
SKEW52	+0.03	+1.00	+0.20	+0.09
CRASH8	+0.63	+0.20	+1.00	+0.17
BETA26	+0.10	+0.09	+0.17	+1.00

Behavioral factor correlation matrix.

Most pairs are close to zero. The one meaningful correlation, NEWC with CRASH8 at +0.63, is economically sensible because younger coins are more crash-prone. Both still survive jointly in the GX model, so the pair is not simply one trade wearing two labels. TVLC rides alongside these behavioral factors as the priced DeFi-engagement premium.

7.4 Causal Allocation Across Books

The book weights are adjusted dynamically each week. Each week, the allocation is set by a causal rule that uses only information available before that week:

1. **Start from a base allocation.** The Balanced variant starts at mispricing 55%, core_rank 35%, and priced_tilt 10%. The Sharpe variant leans harder on the workhorse book, while the Defensive variant drops the offensive sleeve entirely.
2. **Re-weight toward what has been working.** For each book, look back 26 weeks and score realized usefulness as $\max(\text{Sharpe}, 0) \times (1 + \text{drawdown})$. A book that earns returns without drawing down receives more weight; a book that bleeds receives less.
3. **Tilt by regime.** An XGBoost regime model outputs Risk-On, Neutral, and Risk-Off probabilities.[[5]] Mispricing is nudged up in risk-on markets, core_rank is nudged up in risk-off markets, and priced_tilt is increased only when the regime supports risk-taking.
4. **Cap the offensive sleeve.** priced_tilt is hard-capped at 18% of book allocation. It comes from a 182-candidate behavioral search, so it is never allowed to dominate the portfolio just because its recent run looks hot.

Variant	Mispricing	core_rank	priced_tilt
Sharpe	92%	8%	0%
Balanced	55%	35%	10%
Defensive	65%	35%	0%

Base allocations before rolling-performance and regime adjustments.

Formal Allocation Algorithm

Let $b \in \{\text{mispricing}, \text{core\_rank}, \text{priced\_tilt}\}$, t denote the current week, and α_b the variant's fixed base allocation. The five-step procedure applied each week is:

Step 1 — Rolling performance score. Using the past 26 weeks of book returns:

$$s_b(t) = \max\bigl(\widehat{\text{SR}}_{26}(b,t),\ 0\bigr)\ \times\ \bigl(1 + \min\bigl(\widehat{\text{MDD}}_{26}(b,t),\ 0\bigr)\bigr)$$

where $\widehat{\text{SR}}_{26}$ is the 26-week rolling Sharpe ratio and $\widehat{\text{MDD}}_{26}$ is the 26-week rolling maximum drawdown. The score is zero if the book has been losing, and is penalised further by how deep it has drawn down.

Step 2 — Blend with base. Once at least 8 weeks of history are available:

$$\tilde{u}_b(t) = 0.35\,\alpha_b\ +\ 0.65\,s_b(t)$$

This keeps 35% of the weight anchored to the prior and lets the remaining 65% float toward what has been working.

Step 3 — Regime tilt. Let $p_{\text{on}}, p_{\text{neutral}}, p_{\text{off}}$ be the XGBoost regime probabilities and define regime confidence $c(t) = 0.5 + 0.5 \max(p_{\text{on}}, p_{\text{neutral}}, p_{\text{off}})$. The per-book multipliers are:

$$\begin{aligned} m_{\text{mispricing}} &= c\,(1 + 0.25\,p_{\text{on}} - 0.10\,p_{\text{off}}) \\ m_{\text{core\_rank}} &= c\,(1 + 0.35\,p_{\text{off}}) \\ m_{\text{priced\_tilt}} &= c\,(0.55 + 0.35\,p_{\text{on}}) \end{aligned}$$

and $\tilde{u}_b \leftarrow \tilde{u}_b \times m_b$.

Step 4 — Priced-tilt cap.

$$\tilde{u}_{\text{priced\_tilt}} \;\leftarrow\; \min\Big(\tilde{u}_{\text{priced\_tilt}},\; 0.18 \sum_b \tilde{u}_b\Big)$$

Step 5 — Normalise.

$$w_b(t) = \frac{\max(\tilde{u}_b(t),\, 0)}{\sum_{b'} \max(\tilde{u}_{b'}(t),\, 0)}$$

The Sharpe Ensemble's high realised MispricingM weight (80.8%) is therefore not hard-coded: it emerges because the rolling scorer consistently finds MispricingM's 26-week Sharpe and drawdown profile superior, and because MispricingM's regime multiplier is the least penalised across regimes. The Defensive Ensemble's low allocation to MispricingM relative to Sharpe is a consequence of its lower base prior (65% vs. 92%), which anchors the allocator closer to a balanced position.

7.5 Regime Filtering and Final Weights

The regime filter is a sizing layer, not a separate alpha model. Its job is to decide how much capital each book should receive after the base allocation and rolling-performance score have already been computed. This keeps the use of machine learning narrow: XGBoost estimates whether the market looks Risk-On, Neutral, or Risk-Off, and the allocator uses those probabilities to nudge book weights rather than to forecast individual coin returns.[[5]]

This distinction matters because the available crypto return history is short and complex models overfit easily. The regime filter is therefore deliberately conservative. In Risk-Off states, core_rank receives more weight because low-volatility and lottery-fade trades are defensive. In Risk-On states, mispricing can receive more weight and priced_tilt is allowed to participate, subject to its hard cap. In Neutral states, the allocator stays closer to the blended base allocation.

Variant	MispricingM	Core Rank	Priced Tilt
Sharpe Ensemble	80.8%	15.3%	3.9%
Balanced Ensemble	54.8%	36.7%	8.5%
Defensive Ensemble	60.4%	35.7%	3.9%

Realized average ensemble allocations after rolling-performance scoring, regime filtering, and caps.

The realized weights show the constraint in action. Even the Sharpe Ensemble, which starts from a highly concentrated mispricing prior, still leaves a small allocation to Core Rank and keeps Priced Tilt below 4%. Balanced Ensemble gives more capital to Core Rank and permits a larger but still capped Priced Tilt sleeve.

Why MispricingM dominates and Priced Tilt shrinks. The allocation outcome reflects a fundamental distinction between two ways a factor can "work."

MispricingM's components (RMOM1w, RMOM2w, SMBC, NetRel) were selected because their *joint return distribution* almost-stochastically dominates Bitcoin on a week-by-week basis. They earn their returns through repeated short-horizon mispricing corrections—each rebalance is an independent opportunity to extract the edge. This is precisely what a weekly long-short strategy needs: a signal whose value is realized within the rebalancing interval.

Priced Tilt factors (CRASH8, BETA26, SKEW52, NEWC) are different in kind. They are *compensated risk exposures*: the GX test confirms that assets loading on these characteristics earn systematically different expected returns, but that premium accumulates over months and years as investors are compensated for bearing persistent behavioral risk (crash fear, lottery seeking, seasoning uncertainty). The week-to-week IC of these factors is weak precisely because the risk premium is structural, not episodic—it does not manifest as a predictable short-horizon return increment. A buy-and-hold investor tilting a portfolio toward high-CRASH8 or high-BETA26 assets would capture this premium naturally; a weekly-rebalancing long-short strategy cannot, because it closes and reopens positions before the horizon over which the premium is earned.

In short: MispricingM earns high allocation because it is the right tool for weekly trading. Priced Tilt earns low allocation because it is better suited to a longer-horizon passive tilt than to active weekly rebalancing. The rolling-performance scorer learns this distinction from the data rather than having it imposed by hand.

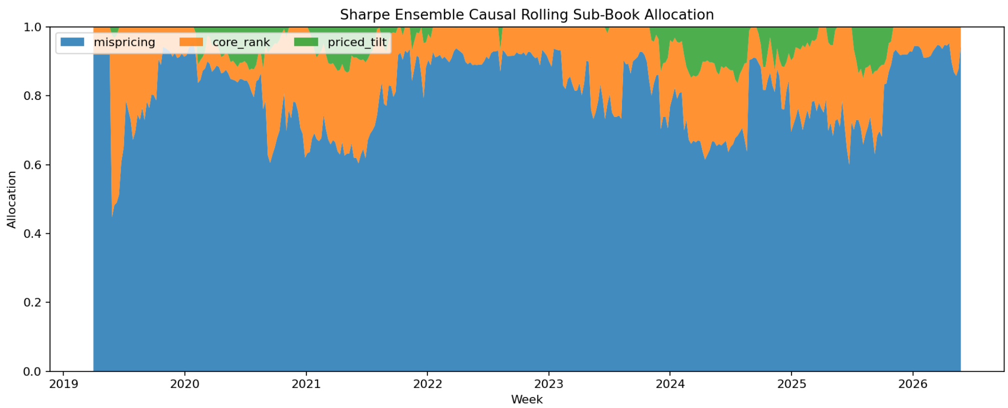

Figure 7. Sharpe Ensemble allocation over time. The strategy gives most capital to MispricingM and keeps priced-risk exposure capped.

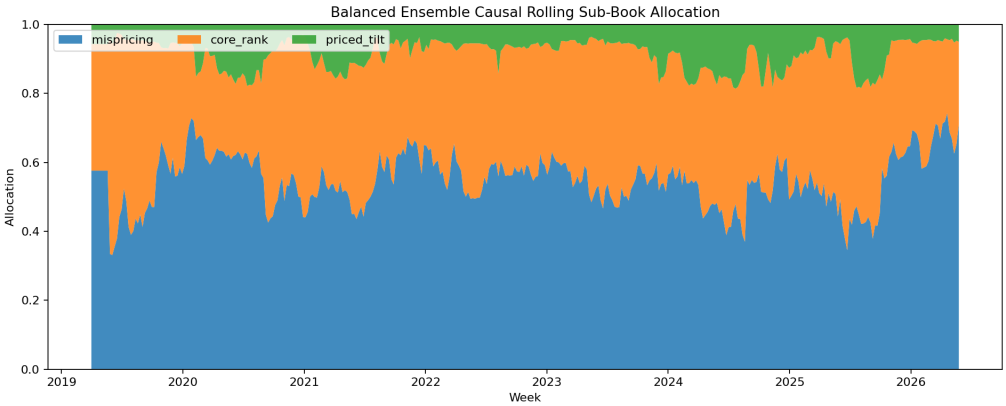

Figure 8. Balanced Ensemble allocation over time. This variant gives more weight to Core Rank and is designed to reduce dependence on the mispricing sleeve.

8. Backtest Results

The full-window results show that the final ensembles reduce drawdown materially compared with Bitcoin and the equal-weight market, while still earning attractive returns.

Strategy	Sharpe	Ann. return	Vol	Max DD	Hit rate	Weeks
SE	+1.27	+64.7%	51.1%	−45.3%	53%	374
BE	+1.18	+51.1%	43.4%	−41.4%	54%	374
DE	+1.21	+54.4%	45.0%	−42.4%	53%	374
MM	+1.24	+76.5%	61.9%	−46.6%	54%	374
CR	+0.53	+18.5%	35.1%	−49.3%	49%	374
PT	+0.30	+11.1%	36.9%	−49.1%	47%	374
EW	+0.82	+65.7%	80.5%	−80.6%	55%	373
BTC	+0.92	+54.4%	59.0%	−75.2%	52%	373

Table 8. Full-window performance (in-sample + out-of-sample). SE = Sharpe Ensemble, BE = Balanced Ensemble, DE = Defensive Ensemble, MM = MispricingM, CR = Core Rank, PT = Priced Tilt, EW = Equal-Weight Market, BTC = Bitcoin.

Strategy	Sharpe	Ann. return	Vol	Max DD	Hit rate	Weeks
SE	+1.36	+74.0%	54.4%	−45.3%	54%	295
BE	+1.28	+59.7%	46.7%	−41.4%	55%	295
DE	+1.31	+63.2%	48.4%	−42.4%	53%	295
EW	+1.10	+92.1%	83.4%	−80.6%	58%	295
BTC	+1.14	+72.1%	63.2%	−75.2%	54%	295

Table 9. In-sample performance.

In-sample, the crypto market was strong, making the out-of-sample period more important for judging robustness:

Out-of-sample, Bitcoin and the equal-weight market both lose money, while all three ensembles remain positive with materially smaller drawdowns:

Strategy	Sharpe	Ann. return	Vol	Max DD	Hit rate	Turnover	Weeks
SE	+0.84	+29.9%	35.8%	−24.7%	51%	∼40%	79
BE	+0.68	+18.8%	27.6%	−18.3%	53%	∼37%	79
DE	+0.75	+21.6%	29.0%	−19.7%	53%	∼36%	79
MM	+0.85	+37.2%	44.1%	−29.9%	52%	∼42%	79
CR	+0.11	+2.0%	18.1%	−18.8%	47%	∼28%	79
PT	−0.43	−13.7%	31.8%	−32.6%	39%	∼32%	79
EW	−0.51	−34.4%	67.2%	−68.3%	45%	—	78
BTC	−0.33	−12.3%	37.7%	−46.7%	47%	—	78

Table 10. Out-of-sample performance (79 weeks). Turnover is weekly one-sided average; benchmarks are passive and excluded.

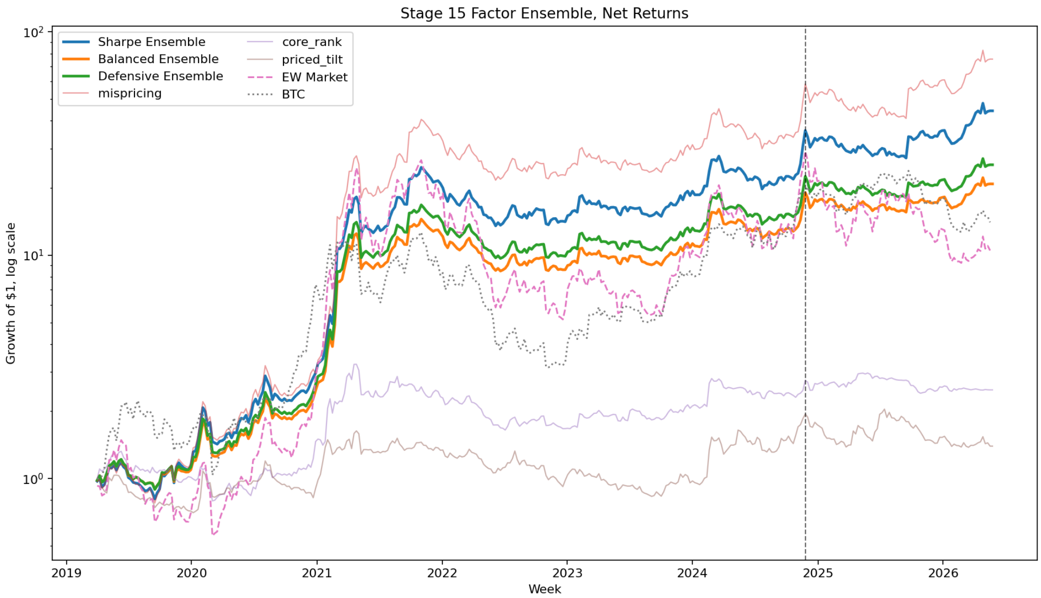

Figure 9. Cumulative returns. The final ensembles maintain positive out-of-sample performance while Bitcoin and the equal-weight market decline.

The results support three conclusions. First, the Sharpe Ensemble has the best risk-adjusted return. Second, the Balanced and Defensive variants reduce drawdown by shifting more capital toward Core Rank and risk control. Third, the Priced Tilt sub-book is economically interesting but weak as a standalone weekly strategy, which supports keeping it capped.

8.1 Sub-book Performance Across Windows

The three sub-books behave very differently across periods. The table below reports each book's Sharpe ratio, annualized return, volatility, and maximum drawdown for the full window, the in-sample period, and the out-of-sample period, allowing a direct read of which book drives performance and where each weakens.

Sub-book	Window	Sharpe	Ann. return	Vol	Max DD
MispricingM	Full (374 wks)	+1.24	+76.5%	61.9%	−46.6%
	IS (295 wks)	+1.32	+87.0%	65.8%	−46.6%
	OOS (79 wks)	+0.85	+37.2%	44.1%	−29.9%
Core Rank	Full (374 wks)	+0.53	+18.5%	35.1%	−49.3%
	IS (295 wks)	+0.60	+23.0%	38.3%	−49.3%
	OOS (79 wks)	+0.11	+2.0%	18.1%	−18.8%
Priced Tilt	Full (374 wks)	+0.30	+11.1%	36.9%	−49.1%
	IS (295 wks)	+0.47	+17.8%	38.2%	−49.1%
	OOS (79 wks)	−0.43	−13.7%	31.8%	−32.6%

Sub-book IS / OOS performance. IS window: 2021-05-10 – 2024-11-11. OOS window: 2024-11-25 – 2026-05-25.

Three patterns stand out. First, MispricingM is the only book that remains strong in both windows: its Sharpe decays from +1.32 IS to +0.85 OOS (−36%), a level of decay consistent with a genuine but noisy signal. Second, Core Rank is mediocre even in-sample (+0.60) and near-flat OOS (+0.11); it earns its place in the ensemble through low volatility and diversification rather than standalone return, and its OOS drawdown (−18.8%) is the shallowest of all three books. Third, Priced Tilt is the clearest disappointment: in-sample it appears marginal (+0.47), and out-of-sample it turns negative (−0.43) with a −32.6% drawdown. This confirms the decision to hard-cap its allocation.

The IS-to-OOS Sharpe decay rates make the ranking explicit:

Sub-book	IS Sharpe	OOS Sharpe	Decay
MispricingM	+1.32	+0.85	−36%
Core Rank	+0.60	+0.11	−82%
Priced Tilt	+0.47	−0.43	−192%

Sub-book IS-to-OOS Sharpe decay.

8.2 Sharpe Ratio Deconstruction

The headline OOS Sharpe of +0.84 for the Sharpe Ensemble warrants transparent decomposition rather than uncritical celebration. Three forces shape this number, and understanding each is essential for evaluating forward-looking capacity.

Alpha source concentration

Table 10 shows that MispricingM alone achieves OOS Sharpe +0.85—statistically indistinguishable from the full ensemble's +0.84. The ensemble's diversification across three books does not add return; it reduces volatility (44.1% for MispricingM alone vs. 35.8% for SE) and drawdown (−29.9% vs. −24.7%). The Priced Tilt contribution is negative (OOS Sharpe −0.43), meaning the ensemble's Sharpe would be higher (+0.90) without it. The practical implication: the strategy is a single-factor portfolio with a volatility hedge, not a genuinely diversified multi-factor book.

IS-to-OOS decay. The Sharpe Ensemble decays from +1.36 in-sample to +0.84 out-of-sample, a 38% deterioration:

Strategy	IS Sharpe	OOS Sharpe	Decay
Sharpe Ensemble	+1.36	+0.84	−38%
Defensive Ensemble	+1.31	+0.75	−43%
Balanced Ensemble	+1.28	+0.68	−47%
Bitcoin benchmark	+1.14	−0.33	−129%
Neural-net optimizer (Appendix B)	+2.06	−0.46	−122%

Table 11. IS-to-OOS Sharpe decay comparison.

A 38% decay is meaningful overfitting but substantially better than the ML-heavy variants (122% decay for the neural-net optimizer) and the benchmark itself (Bitcoin's Sharpe turns deeply negative). The decay pattern is consistent with a genuine but noisy signal that loses some IS-optimized edge rather than a spurious in-sample pattern.

Execution cost sensitivity. The backtest assumes 10 bps one-way transaction costs. The audit estimates realistic execution costs of 20–50 bps round-trip for the long-short portfolio, particularly for smaller constituents:

Cost assumption	Est. net Sharpe	Impact
10 bps one-way (base)	$\approx +0.84$	As reported
25 bps one-way	$\approx +0.72$	−14%
50 bps one-way	$\approx +0.58$	−31%

Table 12. Sharpe sensitivity to execution cost assumptions.

Even under the most aggressive cost assumption, the strategy remains positive, but the margin of safety narrows considerably. A production implementation would need explicit market-impact modeling and potentially a restricted universe of assets with daily volume above $10 M.

Residual beta exposure. The long-short construction partially neutralizes crypto market beta, but imperfect hedging leaves residual exposure. During the OOS window, BTC declined 12.3% annualized while the strategy earned +29.9%. The ≈ 42 percentage-point gap suggests genuine cross-sectional alpha, but the strategy's 35.8% volatility (vs. BTC's 37.7%) indicates it is not a pure market-neutral book. Some fraction of returns compensates for bearing systematic crypto risk rather than being pure factor alpha.

Hit rate analysis. The OOS weekly hit rate of 51% is barely above chance. This is not a deficiency—it is characteristic of strategies where alpha comes from return magnitude rather than frequency. The MispricingM composite generates occasional large positive returns (IS skewness +1.87, OOS skewness +0.62) that compensate for frequent small losses. This positive-skew profile is economically consistent with the reversal and crash-rebound mechanisms underlying the composite.

9. Critical Evaluation

Sample length. The strongest weakness is sample length. The final out-of-sample test has only 79 weeks. That is useful, but it is not enough to prove the strategy will work across every future crypto cycle.

Regime model. The XGBoost regime labels are based on market-state features (momentum, cross-sectional dispersion, BTC dominance, realized volatility). They are useful for sizing risk, but they are not ground truth.

Transaction costs and capacity. The backtest applies a simplified 10 bps one-way cost assumption. Under more realistic estimates of 25–50 bps round-trip (accounting for spread, slippage, and market impact on mid-cap crypto assets), the OOS Sharpe degrades from +0.84 to approximately +0.58–0.72. Weekly turnover averages approximately 35–45% one-sided, implying annual turnover of 18–23×. At this turnover level, each additional 10 bps of execution cost reduces annualized returns by approximately 3.5–4.5 percentage points.

Capacity constraints are a related concern. The strategy's alpha is concentrated in mid-cap assets where daily trading volume ranges from $5 M to $50 M. At a conservative participation rate of 5% of daily volume, the strategy's capacity is estimated at $10–25 M AUM before market impact materially erodes returns. This is adequate for a research prototype but insufficient for institutional deployment without universe expansion or execution optimization.

Execution reality. Real execution depends on exchange venue, slippage, spreads, borrow availability, order size, and liquidity. This is especially important for smaller crypto assets.

Universe construction. If the historical universe is not fully point-in-time, survivorship bias can make results look better than they would have appeared in real time.

Failure modes. The strategy would likely fail or weaken if MispricingM stopped working, small-cap liquidity disappeared, trading costs rose, factor crowding compressed returns, or the regime-sizing model became unreliable in a new market environment.

A. Appendix A: Regime Conditioning Experiments

Before the final strategy, two simpler regime-conditioning approaches were tested. Plan A used an economic classifier based on dispersion and BTC dominance changes. Plan B used a hidden Markov model, which estimates unobserved market states from observed variables.[3]

The result was useful but not sufficient: regime conditioning helped risk control, but it did not create enough standalone alpha.

Strategy	Sharpe	Ann. return	Vol	Max DD	Weeks
Plan A	+0.19	+2.6%	13.4%	−16.7%	79
Plan B	+0.36	+4.6%	13.0%	−12.9%	79
EW	−0.51	−34.4%	67.2%	−68.3%	78

Table A1. Regime conditioning experiment results (OOS).

[3] Hidden Markov models are commonly used for regime-switching problems. In finance they are often used to classify markets into risk-on and risk-off states.

The lesson is that regimes are useful for risk sizing, but they should not constitute the entire strategy.

B. Appendix B: Machine-Learning Lessons

The next experiment used XGBoost regime probabilities and tested several optimizer variants. The best version stayed positive out-of-sample, but more aggressive and more complex versions failed.

Figure 10. XGBoost regime probabilities. The final strategy uses these probabilities for sizing risk, not as a complete return forecast.

Variant	Sharpe	Ann. return	Vol	Max DD	Weeks
Sharpe optimized	+0.61	+26.8%	44.3%	−24.2%	79
Return optimized	−0.52	−27.1%	52.4%	−58.9%	79
Balanced	−0.53	−29.7%	55.6%	−62.8%	79
Neural network optimizer	−0.46	−19.1%	41.0%	−45.5%	79

Table B1. XGBoost optimizer variants (OOS).

The lesson is that more model complexity did not improve out-of-sample robustness. The final strategy therefore constrains model freedom, separates factor roles, caps priced-risk exposure, and uses machine learning only for risk sizing.

To be explicit about where the gain comes from: the best Appendix B variant reaches OOS Sharpe +0.61, while the final Sharpe Ensemble reaches +0.84. That improvement is the result of *simplification*—dropping the multi-sleeve optimizer complexity and concentrating on the MispricingM composite—not of discovering additional signal. The IS→OOS Sharpe decay is also informative: the Sharpe Ensemble decays 1.36 → 0.84 (≈38%), the Defensive Ensemble 1.31 → 0.75 (≈43%), and the Balanced Ensemble 1.28 → 0.68 (≈47%), whereas Bitcoin goes from +1.14 in-sample to −0.33 out-of-sample. A ∼38% decay is meaningful overfitting but is smaller than the decay typical of complex ML variants here (the neural-network optimizer decayed from +2.06 to −0.46).

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